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PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:	February 2021 report: Professor Michael Gollner, Co-PI.		
COI Name (Institution):	Gollner, Michael Ph.D. (University of California, Berkeley)		
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Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. The flammability, or fire hazard, of materials is normally assessed by the material's ease of ignition, rate of flame spread, heat released and extinction. In this research project, an experimental investigation is conducted to study the flammability of thermoplastic materials under SEA is being conducted. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, burning (flame spread and mass burning), and extinction characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
	This Task Progress Report presents a summary of the research conducted during the last year of the current funding period. The research conducted in previous years has been already reported in the corresponding annual research reports. It includes a brief description of the normal gravity MIST experimental apparatus and the results obtained during this period. Experiments The experimental apparatus used in the normal gravity experiments has been described in previous progress reports except for the inclusion of radiant heaters, and thus it will be described here only briefly. It consists of a laboratory scale combustion tunnel that is inserted in a 105 L pressure chamber. The chamber is provided with a flow system that provides constant supply, and exhaust, of gases to avoid vitiation problems. Compressed house air, or nitrogen and oxygen, are supplied through critical nozzles (O'Keefe Controls) to the bottom of the combustion tunnel while constantly evacuating to maintain constant the pressure inside the chamber. The chamber pressure is controlled with a high-capacity vacuum generator (Vaccon JS-300) and a mechanical vacuum regulator. The chamber pressure is monitored constantly with an electronic pressure transducer (Omega Engineering, Inc. PX303-015A5V). Tests can be conducted at sub-atmospheric pressures ranging between 100 kPA and 30 kPa ± 2 kPa, and varied oxygen concentrations. The forced flow velocity can be varied from 300 mm/s. The flow duct was built following a similar setup used to study flame spread over solid fuels together with the details of the heater attachment that will be used in the Solid Fuel Ignition and Extinction / Material Ignition and Suppression Test (SoFIE/MIST) microgravity experiments that are planned to be conducted on board the International Space Station (ISS). The laboratory scale combustion tunnel has a cross sectional area of 125 mm by 125 mm, and is 600 mm in length. The first 350 mm of the duct serves as a flow straightener, where inlet gases pass sequentially through a 20 mm layer of 5 mm borosilicate beads, a stainless-steel mesh and 63 mm thick aluminum honeycomb with 3 mm cells before entering the test section. The last 250 mm portion of the duct is used as the test section. The side walls of the top portion of the test section are made of Pyrex glass to withstand the heat from the heater array and the burning samples, while allowing the visualization of the burning samples. The samples used in the experiments are cylindrical rods made of cast black polymethyl methacrylate (PMMA). The rods are placed vertically in the center of the test section, supported on a metal cylinder of the same diameter to prevent flow disturbances. The black PMMA was selected to maximize the absorptance of the thermal radiation at the rods' surface. The PMMA rods are 90 mm long with diameters of 6.4 mm, 9.5 mm, and 12.7 mm. The sample material and geometry were selected following previous experiments conducted in the ISS under the BASS-II project and future experiments that will be done as part of the SoFIE/MIST project. Also, PMMA is potentially going to be used in some spacecraft components, such as windows, in future space vehicles. The samples are positioned in the center of the test section using two actuators (Firgelli Automations FA-150-S-12-6) situated on opposite sides of the flow straightener section of the duct and powered with a power supply (MASTECH HY1803D). The actuators are connected by a central shaft that holds and positions the sample holder. Several K-type thermocouples are embedded on the surface of the PMMA samples to record surface temperature from the beginning of the test until the moment the flame arrives to the location of the bead. Four flat ceramic heaters (Bach Resistor Ceramics GmbH) are used as the source of external thermal radiation. These heaters were selected to match the heaters that will be used in the MIST/SoFIE experiments. The selection of the heaters for the MIST/SoFIE experiments was made based on concerns that other heaters available, like halogen lamps, may shatter during the launch of the rocket where the MIST apparatus would be transported to the ISS. Each ceramic heater has a length of 75 mm, a width of 14.7 mm, and a heated section of 50 mm. The four heaters are mounted on fixed rails positioned on each one of the corners of the experimental section surrounding the sample. Heaters are equi-spaced with a distance of 29.85 mm from the heater surface to the center of the duct, where the cylindrical sample is positioned.

Task Progress:

Parabolic reflectors made of electropolished stainless steel are positioned behind the heaters with the polished side facing the sample. The reflectors are 70 mm in length by 51 mm in width and they are positioned 18.5 mm from the back of the heater. The heater and reflector are positioned 83 mm and 93.2 mm, respectively, from the top opening of the duct, aligned with the position of the sample. The four heaters are used to provide a variable heat flux in the direction of flame spread (downward). The distribution of heat flux over the solid surface provided by the heaters was characterized under different power settings using a Schmidt-Boelter radiometer from Medtherm Corporation. The characterization of the heaters was performed considering the heaters assembled in the tunnel with the array of reflectors. Distribution in the direction of flame spread is monitored for five different power settings ranging between 30 V and 70 V, with peak heat fluxes ranging from 0.9 to 9.3 kW/m2 respectively. Higher voltage settings are not considered because at those conditions the peak heat flux over the sample would exceed the critical heat flux for ignition of PMMA.

Experimental results

The opposed flow flame spread over the surface of the PMMA rods was investigated under the influence of different levels of radiation. The primary data collected during the experiments was the flame spread rate over the PMMA surface, and the surface temperature at the thermocouples' location. The rate of flame spread was determined by tracking the position of the flame leading edge in the recorded videos on a frame by frame basis with an internally developed Python-based program called Flame Tracker. Once ignition was obtained and the sample loaded into the test section, the flame was observed as it propagated downward over the sample. As the radiant heat flux is increased, the flame spread rate speeds up and the flame burns stronger and brighter. As the PMMA rods burn, a conical shape at the tip of the sample is formed and no dripping of molten plastic is observed. During the burning process, significant vapor jetting of bubbles of PMMA bursting at the surface are also observed, causing distortions in the flame as the test progresses.

The measurements show that as the heat flux is increased the flame spreads more rapidly due to the preheating of the PMMA before the arrival of the flame. Furthermore, the flame has an initial acceleration period (soon after ignition) and a deceleration period (closer to the end of the spread) following the profile of the applied heat flux. From the flame spread data, it is possible to see how the flame spread rate is influenced by the radiant heat flux, resulting in faster flame spread rates for the higher heat fluxes. In addition, the preheating time along the fuel sample at a given location is different for different heat fluxes because of the different flame spread rates.

Data correlation

The behavior of the flame spread over a solid fuel is a process that involves multiple physicochemical processes that affect the gas (transport, mixing, chemical kinetics) and solid (heat transfer, thermal decomposition, gasification) phases. As the flame spreads over the solid surface, enough heat has to be transferred from the flame to the unburnt solid ahead of the flame to raise its temperature to the pyrolysis temperature and pyrolyze it. The pyrolyzed fuel is then convected and diffused away from the surface where it mixes with the oxidizer, forming a flammable mixture that is later ignited by the flame in a continuous process. The spread of the flame is therefore determined by the heat transferred from the flame to the solid, the rate at which the fuel is pyrolyzed and the ability of the flame to ignite the flammable mixture. In the presence of an external source of radiant heating, the external radiant flux contributes in two primary ways to the enhancement of the flame spread rate: on the one hand, the added radiant heat increases the temperature of the unburnt solid in the region ahead of the flame; on the other, the external radiant heat flux also increases the fuel gasification rate in the burning region because of the thermal energy applied to the surface in addition to that provided by the flame.

Despite the complexities involved in the flame spread process, several theoretical models have been developed to simulate and provide a better understanding of the phenomenological process and the effect of different parameters. Most of the different theoretical models for downward flame spread proposed in the literature predict a flame spread rate that is proportional to the external radiant flux and inversely proportional to the difference between the pyrolysis temperature of the solid and the surface temperature of the solid at the time of flame front arrival to a specific location. Consequently, the flame spread rate should correlate with either the solid surface temperature at the time of flame arrival, or with the energy received by the solid at the time of flame arrival. This is described in the following two sections.

Data correlation based on the surface temperature at the moment of flame arrival

Looking at the theoretically derived expression of the flame spread rate, it is seen that the applied external radiant heat flux affects the spread rate in two ways: directly, and indirectly (through variations in the surface temperature beyond the region heated by the flame). With no external radiant flux, the temperature of the surface that the flame experiences can be assumed to be the sample's initial temperature T_o. However, in the presence of an external source of radiant heating, the surface temperature will increase as a function of the magnitude of the heat flux applied and the time that the solid is exposed to the heat flux. For downward/opposed flame spread, the flame does not heat the portion of the solid far ahead of the flame front as it spreads; thus, approximately it can be assumed that T_s is the surface temperature at the time of flame arrival to a particular location t_arr. Accordingly, in an opposed flow configuration under an external heating like the one studied in the present work, the spread process is strongly influenced by the temperature of the fuel, which in turn is changing in time as a result of the applied non-uniform external heating. Therefore, as the flame spreads over the solid, the leading edge of the flame encounters a surface temperature that is changing as a function of time and space, influencing in turn the rate at which the flame spreads.

Following this concept, the measured flame spread velocity was correlated in terms of a nondimensional temperature at the time of flame arrival. It was found that the flame spread rate measured correlates fairly well with the surface temperature, with larger spread rates following larger surface temperatures. The good agreement of the correlation is a statement of the importance of the surface temperature, and in turn the external radiant heat flux, in determining the rate of spread of the flame.

Data correlation based on the energy applied to the surface prior to flame arrival

Although useful and widely used in the literature, thermocouples present some limitations when used to measure the surface temperature of the solid (i.e., they need to be carefully attached to the surface to minimize inconsistencies, they are susceptible to heating from the radiant flux and signal noise, they are labor intensive, among others). Furthermore, thermocouple measurements will not be available during the flight experiments performed as part of the SoFIE/MIST experiments. Thus, additional methods to determine the surface temperature, and to correlate the flame spread rate at different environmental conditions, need to be explored.

For downward flame spread over a surface exposed to external radiant heating, the applied external heat flux determines

	the amount of energy that the sample receives as preheating while the flame is spreading. Consequently, the surface temperature that the flame experiences at the moment of arrival to a given location in the sample is also determined by the energy received by the sample at that location and time. Given the complexities associated with temperature measurements, and the limited accessibility for those type of measurements in future spacecraft experiments, an alternative approach is pursued here with the purpose of correlating the flame spread rate with the energy levels applied to the surface. The reasoning behind this approach relies on the idea that the surface temperature is determined by the balance between the energy applied at the surface and the energy removed from it due to heat losses. For a flame spreading in a downward configuration away from extinction conditions, the energy applied to the surface is larger than the energy losses, and consequently the spread process develops in a continuous fashion. In the presence of an external radiant heat source, the total energy applied to the surface becomes dominant over the energy losses and thus it is expected that the flame spread rate could be directly correlated to this parameter, as it was previously done with the surface temperature. The energy values are obtained from the relation between the external heat flux applied to the surface and the time the sample is exposed to the applied heat flux (tarr). Plots of the surface temperature and energy received by the solid at the time of flame arrival and energy are then compared; as expected, both the spatial and temporal profiles of the energy applied to the surface distributions follow similar distributions as those of the heater calibration, sample surface temperature at the time of flame arrival, and the flame spread rate. The similarities between these distributions are a clear demonstration of how intrinsically related the spread rate is to the surface temperature and to the energy applied to the surface. The measured flame spread velocity was plotted in a nondimensional form as a function of a nondimensional energy received by the sample at the location and time of flame arrival. Flame spread rate and energy are nondimensionalized using the average flame spread rate obtained under no external heat flux, V_(o,exp), and the energy required to raise the surface temperature to the pyrolysis temperature, respectively. The energy applied to the surface is determined as a function of the time it takes the flame to reach a specific location (t_arr) and the external heat flux applied to the surface. It was found that the measured flame spread rate correlates well with the energy, with larger spread rates following larger energies. These results provide an alternative method to correlate the flame spread rate under variable radiant heat fluxes, based only on the energy applied to the surface. The use of the correlations presented here removes the need to measure the surface temperature, and thus facilitates the analysis of data in which the means to constantly measure surface temperature are not available. A paper reporting these results has been accepted for publication in Fire Technology.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Papers from Meeting Proceedings	Thomsen M, Fereres S, Carmignani L, Fernandez-Pello C, Ruff GA, Urban DL. "Determining the cause of reduced concurrent flame spread over thin solid fuels in low pressure and low gravity." 50th International Conference on Environmental Systems - ICES 2021, Virtual, July 12-15, 2021. 50th International Conference on Environmental Systems - ICES 2021, Virtual, July 12-15, 2021. ICES paper ICES-2021-102. , Jul-2021